Medical diagnosis system using supervised classification

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Running title: Medical diagnosis system using supervised classification

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ABSTRACT: Chronic disease prediction from population health surveys enables targeted preventive interventions. This study develops a multi-disease prediction pipeline using the 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset, comprising 441,456 respondents and 330 features collected across the United States. A 10% stratified random sample (44,146 records) is used for analysis. Eleven chronic conditions are modeled as binary classification targets, including diabetes, coronary heart disease, stroke, depression, arthritis, COPD, and asthma. A curated set of 15 demographic and lifestyle features is used after ANOVA F-score selection, with SMOTE applied to address class imbalance. Six classifiers are trained and evaluated: Logistic Regression, Support Vector Machine (RBF kernel), Random Forest, Multi-Layer Perceptron, Gradient Boosting, and Naive Bayes. For the primary target (diabetes), Gradient Boosting achieves the highest AUC-ROC of 0.9260, with Random Forest attaining 94% accuracy. Across 11 diseases, diabetes achieves the highest AUC (0.9138) while asthma is the most challenging (0.6144). Results demonstrate the feasibility of survey-based machine learning pipelines for population-level chronic disease screening.

Key words: chronic disease prediction, BRFSS, machine learning, random forest, gradient boosting, multi-disease classification

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1. INTRODUCTION

Chronic non-communicable diseases such as diabetes, cardiovascular disease, and depression account for the majority of global mortality and healthcare expenditure. The Behavioral Risk Factor Surveillance System (BRFSS) is a United States-based telephone health survey collecting data on health behaviors, chronic conditions, and preventive service utilization from over 400,000 adult respondents annually. Machine learning models trained on BRFSS survey responses can enable cost-effective screening by identifying high-risk individuals from routinely collected lifestyle and demographic information, without requiring expensive diagnostic tests.

This study implements a complete multi-disease prediction pipeline on the BRFSS 2015 dataset, targeting 11 chronic conditions simultaneously. The objectives are: (i) to identify the most predictive lifestyle and demographic features for chronic disease; (ii) to compare six classifiers on the primary target of diabetes prediction; and (iii) to evaluate generalization of the best-performing classifier across all 11 disease targets.

2. DATASET AND PREPROCESSING

2.1. BRFSS 2015 Dataset

The BRFSS 2015 dataset contains 441,456 respondent records with 330 features covering demographics (age, sex, income, education), lifestyle behaviors (BMI, physical activity, sleep, alcohol, smoking), and self-reported chronic condition diagnoses. Eleven binary disease targets are modeled: heart attack (CVDINFR4), coronary heart disease (CVDCRHD4), stroke (CVDSTROKE3), skin cancer (CHCSCNCR), other cancer (CHCOCNCR), COPD (CHCCOPD1), arthritis (HAVARTH3), depression (ADDEPEV2), kidney disease (CHCKIDNY), diabetes (DIABETE3), and asthma (ASTHMA3). A 10%

stratified random sample (44,146 records) is used throughout to balance computational feasibility with statistical representativeness.

Figure 1 is about here.

2.2. Feature Selection and Engineering

From 330 raw features, 15 clinically motivated variables are selected: age (_AGE80), sex, BMI (_BMI5), general health (GENHLTH), physical health days (PHYSHLTH), mental health days (MENTHLTH), sleep hours (ADSLEEP), physical activity (EXERANY2), smoking status (SMOKE100), heavy drinking (DRNKANY5), income level (INCOME2), education (EDUCA), healthcare access (HLTHPLN1 and MEDCOST), and employment status (EMPLOY1). Class imbalance is severe for all disease targets; SMOTE oversampling is applied to the training set to balance classes before model fitting. The final working sample after SMOTE for the diabetes target contains 9,380 balanced training instances.

Figure 2 is about here.

3. METHODS

Six classifiers are trained and evaluated on the diabetes target: Logistic Regression (L2, max_iter = 1000), SVM with RBF kernel (C = 1, gamma = 'scale'), Random Forest (100 estimators), Multi-Layer Perceptron (hidden layers: 100, 50), Gradient Boosting (100 estimators, learning_rate = 0.1), and Gaussian Naive Bayes. An 80/20 stratified train-test split is applied before SMOTE, which is fitted only on training data to prevent leakage. Performance is measured by accuracy, AUC-ROC, F1 score, and per-class precision and recall. PCA and t-SNE visualizations assess feature-space separability. The best classifier is then applied independently to all 11 disease targets to assess generalization.

Figure 3 is about here.

4. RESULTS

4.1. Disease Prevalence and Class Imbalance

Disease prevalence varies substantially: arthritis is most prevalent (~33%), followed by depression (~20%) and diabetes (~15%). Cardiovascular conditions (heart attack, stroke) have prevalence below 5%, creating severe class imbalance requiring SMOTE. Figure 1 shows prevalence distribution and Figure 2 illustrates class imbalance severity across all targets.

4.2. Model Comparison — Diabetes Target

Table 1 presents results for all six models on the diabetes test set. Gradient Boosting achieves the highest AUC-ROC (0.9260) while both Random Forest and Gradient Boosting reach 94% accuracy. Logistic Regression attains the highest minority-class recall (0.86) at the cost of lower accuracy (0.82). Naive Bayes achieves high minority recall (0.96) but very low accuracy (0.49) due to excessive false positives. ROC curves (Figure 4) confirm ensemble method dominance, and confusion matrices (Figure 5) illustrate the precision-recall trade-off across models.

Table 1 is about here.
Figure 4 is about here.
Figure 5 is about here.

4.3. Multi-Disease Generalization

Table 2 and Figure 6 present AUC-ROC results for Random Forest applied across all 11 disease targets. Diabetes achieves the highest AUC (0.9138), benefiting from strong behavioral correlates with BMI, age, and physical activity. COPD (0.787) and depression (0.772) also achieve competitive discrimination. Asthma (0.614) and other cancer (0.680) are the most challenging targets due to weaker correlations with the selected lifestyle features.

Table 2 is about here.

Figure 6 is about here.

5. DISCUSSION

Gradient Boosting and Random Forest emerge as the strongest classifiers for BRFSS-based prediction, achieving AUC-ROC above 0.91 for diabetes. The dominance of age, BMI, general health, and physical health days as predictors is consistent with established epidemiological evidence. Low precision for the No Disease class across all models reflects the difficulty of identifying the healthy minority after SMOTE balancing. Asthma (AUC 0.614) and other cancer (0.680) are the most challenging targets due to weaker lifestyle correlates in the selected feature set. Future work should address probability calibration, fairness across demographic subgroups, and richer feature sets including medication history and comorbidity indicators.

6. CONCLUSION

A multi-disease machine learning pipeline trained on the BRFSS 2015 survey achieves AUC-ROC of 0.926 for diabetes and above 0.75 for seven of eleven chronic conditions. Gradient Boosting is the best classifier overall; Random Forest provides the best accuracy-generalization balance across all disease targets. Age, BMI, general health, and physical activity are consistently the most predictive features. The study confirms that population health surveys can support scalable machine learning-based chronic disease screening as a cost-effective complement to clinical diagnostics.

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Fig. 6. Multi-disease AUC-ROC results for Random Forest across all eleven BRFSS 2015 disease targets.

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TABLE

Table 1. Classification results for six models on the BRFSS 2015 diabetes prediction target

Model	Accuracy	AUC-ROC	F1 (Disease)	Recall (No Dis.)	Prec.
Logistic Regression	0.82	0.923	0.89	0.86	0.23
SVM (RBF)	0.87	0.916	0.93	0.77	0.29
Random Forest	0.94	0.914	0.97	0.49	0.53
MLP	0.91	0.873	0.95	0.50	0.33
Gradient Boosting	0.94	0.926	0.97	0.64	0.47
Naive Bayes	0.49	0.859	0.63	0.96	0.10

Green rows = best accuracy models. Prec. = precision for No Disease class. Training set after SMOTE: 9,380 balanced instances; Test set: 1,246 instances.

Table 2. Multi-disease AUC-ROC results for Random Forest across all eleven BRFSS 2015 disease targets

Disease	AUC-ROC (Random Forest)
Diabetes (DIABETE3)	0.9138
Arthritis (HAVARTH3)	0.7651
Depression (ADDEPEV2)	0.7722
COPD (CHCCOPD1)	0.7869
Heart Attack (CVDINFR4)	0.7566
Coronary Heart Disease (CVDCRHD4)	0.7536
Stroke (CVDSTROKE3)	0.7203
Skin Cancer (CHCSCNCR)	0.7235
Other Cancer (CHCOCNCR)	0.6798
Kidney Disease (CHCKIDNY)	0.7000
Asthma (ASTHMA3)	0.6144

Green row = highest AUC. Feature set: 15 demographic and lifestyle variables; SMOTE applied per target.

Fig. 1. Disease prevalence for all eleven chronic condition targets in the BRFSS 2015 dataset.

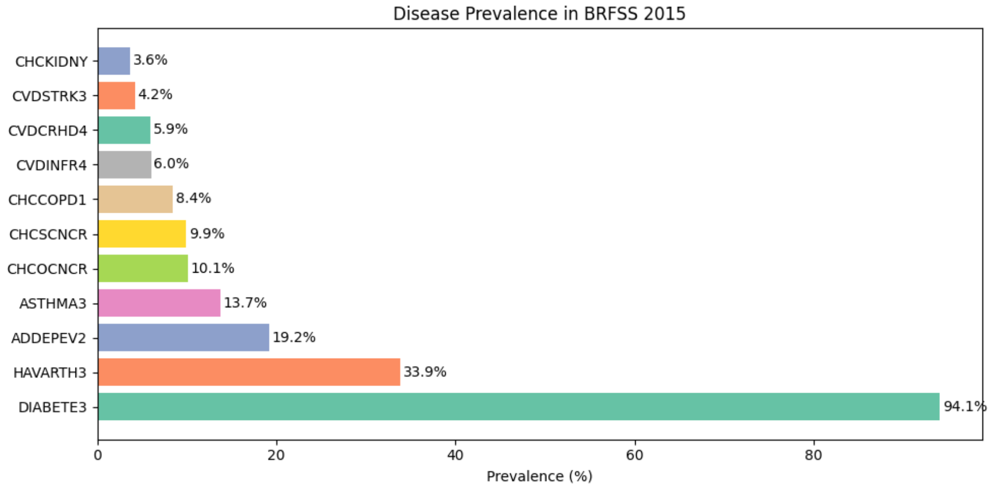


Fig. 2. Class imbalance across all eleven disease targets showing disease vs. no-disease ratios.

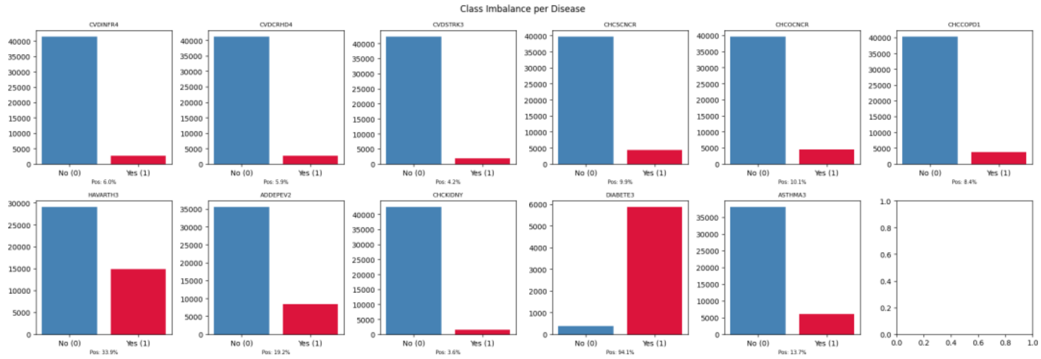
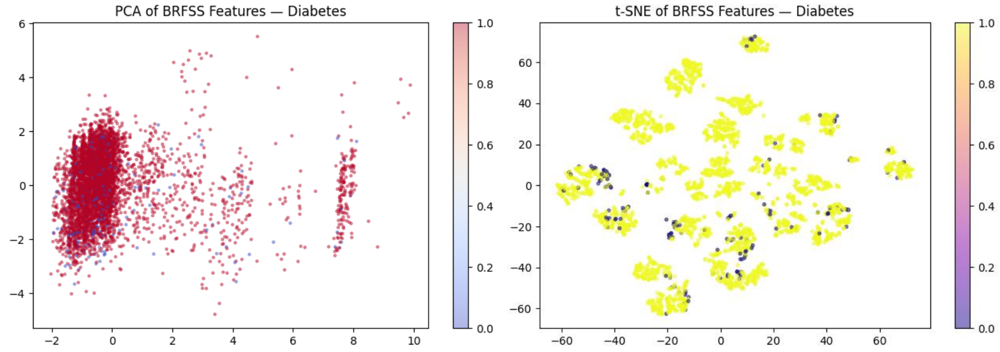


Fig. 3. PCA (left) and t-SNE (right) two-dimensional projections of the selected feature space for the diabetes target.



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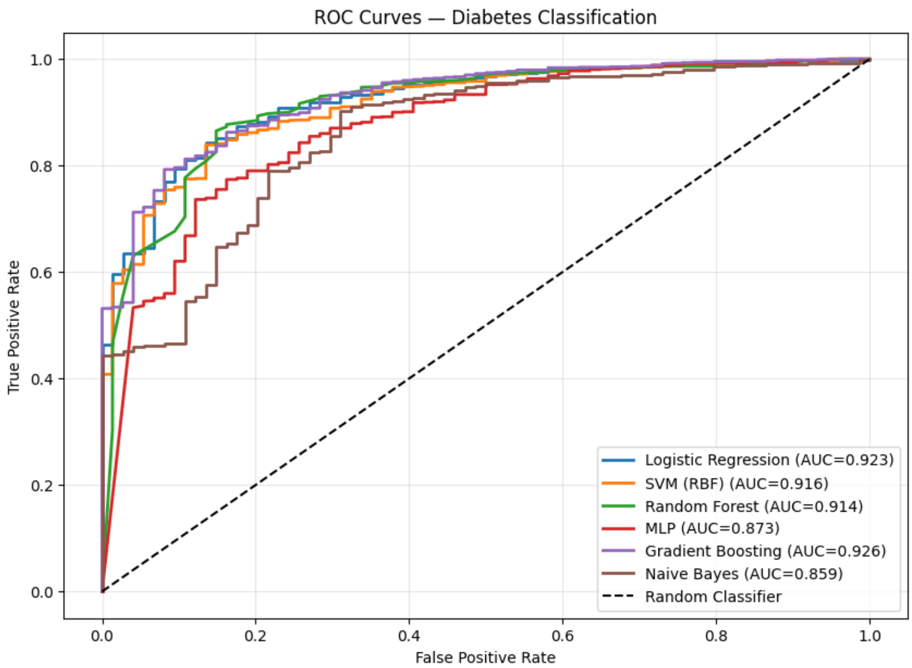

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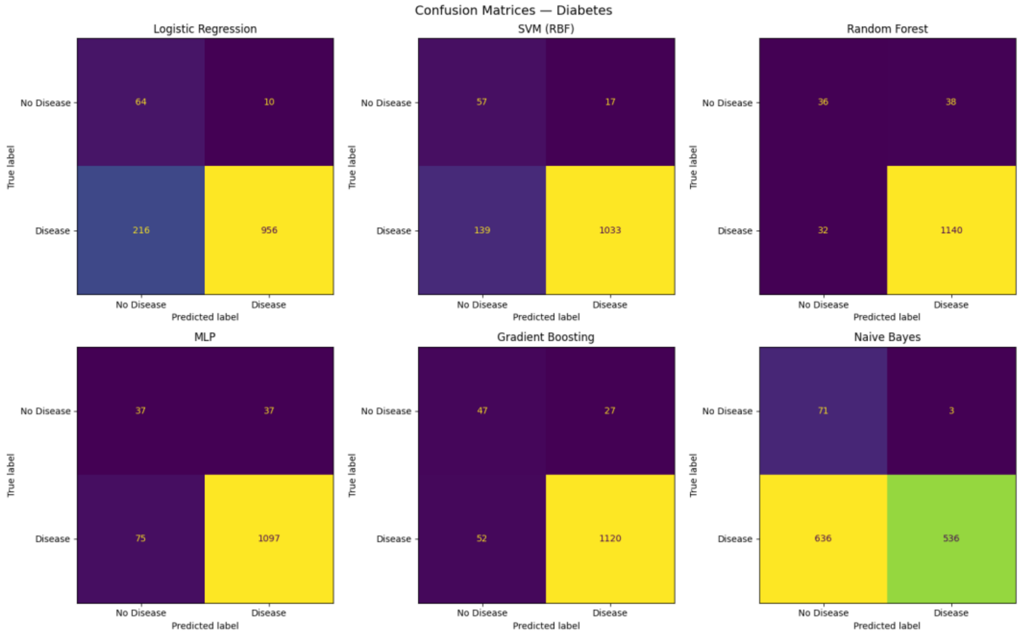

Fig. 6. Multi-disease AUC-ROC results for Random Forest across all eleven BRFSS disease targets.